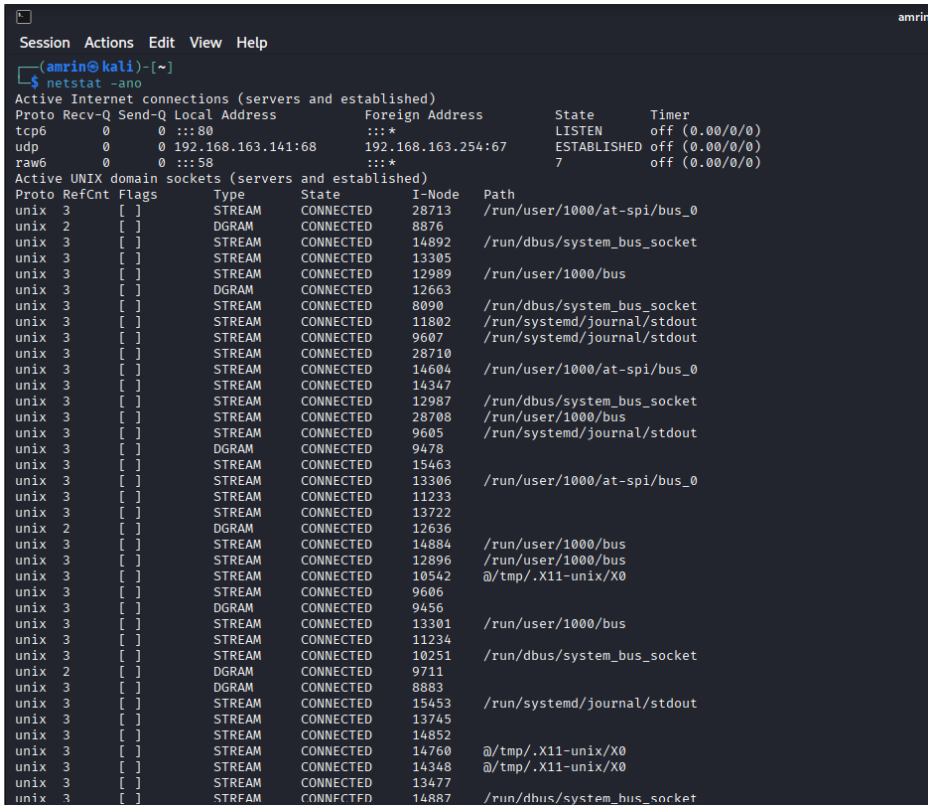
1.Use the netstat command on your Windows or Linux system to list all currently open ports and their associated services, then take a screenshot of the output.

Ans-



```
Session  Actions  Edit  View  Help
┌──(amrin㉿kali)-[~]
└─$ netstat -ano
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       Timer
tcp6       0      0 :::80                   :::*                    LISTEN      off (0.00/0/0)
udp        0      0 192.168.163.141:68      192.168.163.254:67      ESTABLISHED off (0.00/0/0)
raw6       0      0 :::58                   :::*                    7           off (0.00/0/0)
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     28713    /run/user/1000/at-spi/bus_0
unix  2      [ ]         DGRAM      CONNECTED     8876
unix  3      [ ]         STREAM     CONNECTED     14892    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     13305
unix  3      [ ]         STREAM     CONNECTED     12989    /run/user/1000/bus
unix  3      [ ]         DGRAM      CONNECTED     12663
unix  3      [ ]         STREAM     CONNECTED     8090     /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     11802    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     9607     /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     28710
unix  3      [ ]         STREAM     CONNECTED     14604    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     14347
unix  3      [ ]         STREAM     CONNECTED     12987    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     28708    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     9605     /run/systemd/journal/stdout
unix  3      [ ]         DGRAM      CONNECTED     9478
unix  3      [ ]         STREAM     CONNECTED     15463
unix  3      [ ]         STREAM     CONNECTED     13306    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     11233
unix  3      [ ]         STREAM     CONNECTED     13722
unix  2      [ ]         DGRAM      CONNECTED     12636
unix  3      [ ]         STREAM     CONNECTED     14884    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     12896    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     10542    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     9606
unix  3      [ ]         DGRAM      CONNECTED     9456
unix  3      [ ]         STREAM     CONNECTED     13301    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     11234
unix  3      [ ]         STREAM     CONNECTED     10251    /run/dbus/system_bus_socket
unix  2      [ ]         DGRAM      CONNECTED     9711
unix  3      [ ]         DGRAM      CONNECTED     8883
unix  3      [ ]         STREAM     CONNECTED     15453    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     13745
unix  3      [ ]         STREAM     CONNECTED     14852
unix  3      [ ]         STREAM     CONNECTED     14760    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     14348    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     13477
unix  3      [ ]         STREAM     CONNECTED     14887    /run/dbus/system_bus_socket
```

```
┌──(amrin㉿kali)-[~]
└─$ netstat -abno
netstat: invalid option -- 'b'
usage: netstat [-vWeenNcCF] [<Af>] -r         netstat {-V|--version|-h|--help}
       netstat [-vWnNcaeol] [<Socket> ...]
       netstat { [-vWeenNac] -i | [-cnNe] -M | -s [-6tuw] }

        -r, --route              display routing table
        -i, --interfaces         display interface table
        -g, --groups             display multicast group memberships
        -s, --statistics         display networking statistics (like SNMP)
        -M, --masquerade         display masqueraded connections

        -v, --verbose            be verbose
        -W, --wide               don't truncate IP addresses
        -n, --numeric            don't resolve names
        --numeric-hosts          don't resolve host names
        --numeric-ports          don't resolve port names
        --numeric-users          don't resolve user names
        -N, --symbolic           resolve hardware names
        -e, --extend             display other/more information
        -p, --programs           display PID/Program name for sockets
        -o, --timers             display timers
        -c, --continuous         continuous listing

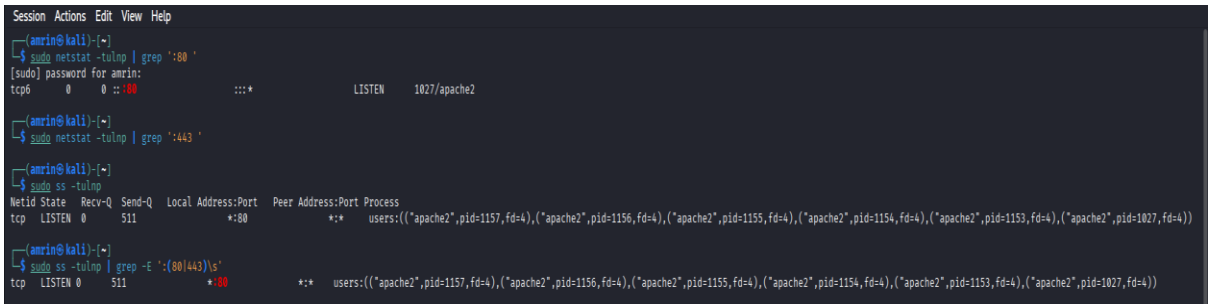
        -l, --listening          display listening server sockets
        -a, --all                display all sockets (default: connected)
        -F, --fib                display Forwarding Information Base (default)
        -C, --cache              display routing cache instead of FIB
        -Z, --context            display SELinux security context for sockets

  <Socket>={-t|--tcp} {-u|--udp} {-U|--udplite} {-S|--sctp} {-w|--raw}
           {-x|--unix} --ax25 --ipx --netrom
  <AF>=Use '-6|-4' or '-A <af>' or '--<af>'; default: inet
  List of possible address families (which support routing):
    inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
    netrom (AMPR NET/ROM) rose (AMPR ROSE) ipx (Novell IPX)
    ddp (Appletalk DDP) x25 (CCITT X.25)

┌──(amrin㉿kali)-[~]
└─$
```

2.Check which applications are using ports 80 and 443 on your system by using the command line tool (netstat, lsof, or Resource Monitor), and write down the process names you find.

Ans-



**3.Scan your own system for open ports using Nmap or Zenmap and identify at least three services running on those ports.

Hint: If you don't have Nmap installed, download it from nmap.org or use Zenmap GUI for easier results.**

Ans-



4.Pick any app you use daily (like Spotify, WhatsApp Desktop, or a local web server) and check which ports it opens on your system while running. List the port numbers and explain in one line what each port is typically used for.

Ans-

Step 1: Open the application

1. Launch Spotify.
2. Play a song or browse the app for a minute.
3. Keep Spotify running while checking its network connections.

Step 2: Find Spotify's process ID

Open PowerShell and run:

Get-Process Spotify |

Select-Object ProcessName, Id

Note the PID values. Spotify may have multiple processes.

Step 3: Find its TCP connections

Run:

$pids = (Get-Process Spotify).Id

Get-NetTCPConnection |

Where-Object { $_.OwningProcess -in $pids } |

Select-Object LocalAddress, LocalPort, RemoteAddress, RemotePort, State

This lists TCP connections associated with the Spotify processes that are currently present. If nothing appears, retry while Spotify is actively playing music.

Step 4: Check UDP connections, if needed

Some applications use UDP, including QUIC over UDP port 443. You can inspect available UDP endpoint information with:

Get-NetUDPEndpoint |

Where-Object { $_.OwningProcess -in $pids } |

Select-Object LocalAddress, LocalPort, OwningProcess

This shows local UDP endpoints, not necessarily every remote UDP connection.

Step 5: Record your observations

Use this table in your practical report and fill it with the ports you actually observe.

Port number	Protocol	Typical purpose
443	TCP	Encrypted HTTPS communication
443	UDP	QUIC-based encrypted communication, if used
Other observed port	TCP/UDP	Explain based on the observed service or connection